

## Making electrical contact to single molecules

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# Making electrical contact to single molecules

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**Abstract.** We are interested in the characterization of electron transport phenomenon through single (bio)molecules. A prerequisite for electrical measurements is the contacting of molecules in the nanometer range. This paper describes various steps toward this goal, starting from statistically distributed molecules (namely, microtubules) adsorbed on a microstructured surface. Monitored by scanning force microscopy and video-enhanced microscopy, adsorption parameters were studied and optimized, including protein-mediated adsorption, induced orientation, and controlled desorption. Line structures connecting the adsorbed molecules to prestructured electrodes were created by means of electron beam-induced deposition (EBD) in a scanning electron microscope. It is shown how such EBD lines can be used as masks for the structuring of metal contacts, and an electrical characterization of these contacts is given.

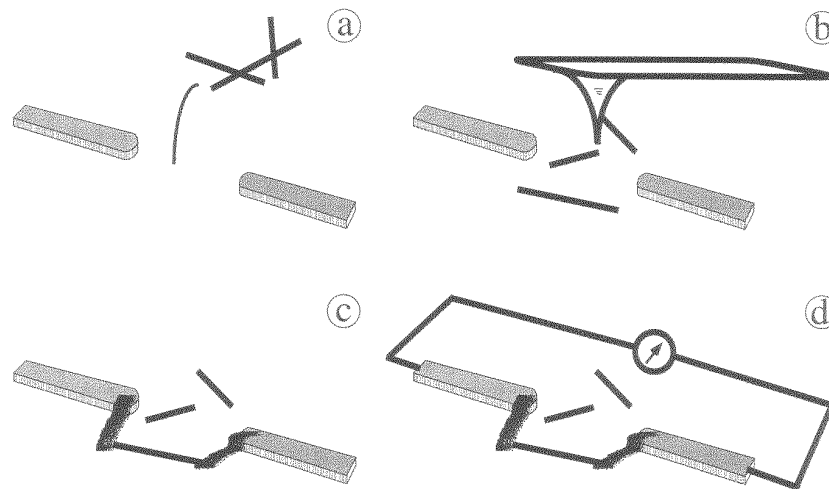
## 1. Introduction

The characterization of the electron transport through single molecules is of great interest for a variety of fields, reaching from microelectronics to biology. Methods for single-molecule detection and manipulation have been established, e.g. based on scanning probe techniques or optical tweezers. An interesting property of molecules is the electron transport mechanism. First studies demonstrated the possibility of contacting and subsequent electrical measurement of single molecules with nanometer dimensions (Bockrath *et al* 1997, Dai *et al* 1996). Approaches in this direction are usually based on scanning probe techniques, due to the random distribution of the molecules over the surface. These techniques are used for electrical measurements (which are hampered by the complex behavior of the tip–sample contact), and/or for detection of the molecules. The detection of the molecules in the scanning force microscope (SFM) is followed by the structuring of electrodes (e.g. using electron beam lithography). Therefore, the basic scheme of a single molecule measurement consists of four steps (figure 1). In the first step, the molecules of interest are adsorbed onto the isolating substrate, usually from a liquid phase (figure 1(a)). The substrate is often already equipped with some electrical contacts. The second step includes the visualization of the molecule distribution, mostly by SFM (figure 1(b)). In the third step the molecule is contacted, either through the conductive scanning tip or by (micro)structuring of electrodes (figure 1(c)). If

a structuring step is involved, the images from SFM are usually used for localization of the molecules of interest. Finally, the fourth step includes the electrical characterization (e.g. current–voltage, figure 1(d)).

This paper presents methodological developments which are mainly focused on the first (adsorption) and the third (contacting) step of the above-mentioned scheme. It describes experiments with microtubules (MTs) as model molecules. MTs are intracellular proteinaceous elements with essential functions in cell architecture and cellular transport, with dimensions in the micrometer (length) and nanometer (width) range. They are usually purified from brain homogenates by temperature-dependent cycles of MT disassembly/reassembly. This procedure yields beside the tubulin, which is the major MT component, some additional proteins, the so-called MT-associated proteins (MAPs). MAPs are bound by basic domains to those sites of the negatively charged tubulin which are located on the MT surface. MTs formed from pure tubulin without MAPs do not bind to a glass surface, which also exposes negative charges on their surface. MAPs, especially the high molecular weight MAP2 ( $M \sim 280$  kDa), known to be responsible for interaction between single MTs as well as between MTs and other cell organelles, have been found to be able to cause also the adsorption of MTs to glass surfaces. Therefore, MAP2 has been used in this study as a suitable tool to adsorb MTs to the substrate.

Line structures between the electrodes and the biomolecules were created using electron beam-induced deposition. These line structures can be transformed into



**Figure 1.** Scheme of a typical experiment for electrical characterization of a single molecule. (a) Adsorption of molecules on prestructured substrates. (b) Visualization of the adsorbed molecules, usually by SFM. (c) Structuring of connective structures between electrodes and molecules, using electron beam-based methods. (d) Electrical characterization.

metal contacts by masking an underlying thin gold film in a dry-etching step.

## 2. Materials and methods

### 2.1. Substrates

Arrays of electrode pairs were structured from  $\sim 100$  nm gold (primed with a  $\sim 5$  nm titanium layer) on a thermally oxidized silicon wafer according to standard photolithographic techniques. The pairs had contact pads of  $400 \times 200 \mu\text{m}^2$  (for convenient contacting by macroscopic electrodes) connected to electrodes of  $1 \mu\text{m}$  width, which were separated by a  $12 \mu\text{m}$  gap (cf figure 4).

For adsorption experiments test structures of gold lines on oxidized silicon wafers with comparable structure sizes were used.

### 2.2. Microtubules

MTs were purified from brain by three cycles of temperature-dependent disassembly/reassembly as described (Shelanski *et al* 1973, Vater *et al* 1983). Using phosphocellulose column chromatography (Weingarten *et al* 1976), the MAPs were separated from the tubulin, which represents the main protein of MTs. The MAP2 was obtained from the pool of MAPs by elution from phosphocellulose with  $1 \text{ M NaCl}$ , followed by heat treatment and gel filtration. In the presented study, MTs were assembled from tubulin both with and without MAPs and stabilized by  $20 \text{ mM taxol}$ .

For adsorption experiments, the assemblies were applied to a substrate of microstructured gold lines on oxidized silicon.

Two different methods of application were used to study the orientation of the MTs. The first method ('running droplet') included the repeated application of droplets on the tilted ( $\sim 45^\circ$ ) chip, so that the droplets were running over and leaving the surface without staying. The other

method ('staying droplet') applied a droplet on a laying chip, followed by an incubation for 5 min. In both cases, the chips were then washed in buffer and aqueous taxol-solution followed by air-drying.

The influence of MAPs on the adsorption properties of MTs was investigated using video-enhanced microscopy. For this purpose, a solution of MAP2 was filled into a glass flow chamber (about  $20 \text{ mm}$  length,  $2 \text{ mm}$  width,  $0.2 \text{ mm}$  height). After adsorption of MAP2 to the glass and washing, a region of the surface was imaged and a suspension of MAP-free MTs was injected under simultaneously microscopic control. The reversibility of MT adsorption was checked by washing the chamber in taxol-containing buffer with high ionic strength ( $0.5 \text{ M NaCl}$ ), breaking open MAP-tubulin binding (Vallee 1983). For contacting, MTs with MAPs were adsorbed to the microstructured gold electrodes described above.

### 2.3. Microscopy

Scanning electron images of uncoated samples were obtained by a digitized scanning electron microscope (SEM) DSM 960 (Zeiss). For SFM a NanoScope III (Digital Instruments (DI), Santa Barbara, CA, USA) with a Dimension 3000 or Multimode head (with lateral scan ranges of  $\sim 130 \mu\text{m}$ ) was used. The microscopes were operated in the tapping mode using tapping mode etched silicon tips (DI).

MT adsorption onto glass was studied by video-enhanced differential interference contrast microscopy, using a microscope Axiophot (Zeiss) equipped with the image processing system Argus 50 (Hamamatsu). Image processing was performed following instructions in (Allen *et al* 1981). The directions of adsorbed MTs were determined from scanning force micrographs after import in the program NIH Image 1.61 (developed at the US National Institutes of Health and available on the Internet at <http://rsb.info.nih.gov/nih-image/>). The fast scan direction (left to right) is thereby assigned as  $0^\circ$ , from which the measured angles opened anticlockwise.

## 2.4. EBD-line structures and gold contacts

For deposition of electron beam-induced deposited (EBD) lines the slow scan axis of the SEM was disabled usually for 20 min. The repeated scanning results in the growth of an amorphous carbonaceous film by polymerization of surface adsorbates (Broers *et al* 1976). This process is limited by surface diffusion. The deposition rate depends on the total  $e^-$ -beam current density at the beam spot (and the rate of emission of secondary and backscattered electrons), on the effective adsorption rate onto the substrate, and on the probability for an electron to cause EBD of a molecule (Folch *et al* 1996). For using the EBD lines as masks a thin layer of gold (about 30 nm) was sputtered onto the sample prior to the EBD structure. The gold was removed by dry etching using Ar ions. Monitoring the electrical resistivity yielded the completeness of the gold removal (by measuring the substrate) as well as the masking efficiency of the EBD lines (by measuring control structures of EBD lines).

## 3. Results

### 3.1. MT adsorption onto microstructured substrates

The adsorption of MTs both with and without MAPs onto microstructured substrates was studied using SFM. Therefore, substrates with gold and silicon oxide surfaces were incubated with assembly solutions, prior to washing and air-drying. The SFM investigation revealed that no adsorption takes place in the case of MTs without MAPs. Only after the addition of MAP2 to the assembly solution, the samples exhibit rod-like structures of some micrometer length, which is the typical appearance of MTs (cf figures 3 and 5). Height measurements yielded 7–10 nm, but were hampered by a surface roughness of up to 3 nm.

### 3.2. MT substrate binding by pre-adsorption of MAP2

The mechanism of substrate binding of MT-MAP2 complexes was investigated by incubating the surfaces in MAP solution, washing and addition of MTs. SFM images of gold and silicon oxide surfaces with MAP2 revealed an increased surface roughness due to the occurrence of globular structures with diameters of  $\sim 25$  nm (measured as height) in a high density (up to 15 per  $\mu\text{m}^2$ ). The addition of MTs to such surfaces resulted in MT-adsorption, comparable to samples of MT-MAP2 complexes. Again, the sample surface was contaminated by the described globular structures.

### 3.3. Reversible attachment of MTs

The reversible attachment of MTs formed from pure tubulin to MAP2-covered glass surfaces was investigated by video-enhanced microscopy. Glass slides incubated with MAP2 revealed no particular structure comparable to the results from SFM of MAP2-coated surfaces. A region with three distinct features (marked by arrowheads in figure 2) was chosen for further observation. After inflow of the MT suspension adsorbed MTs became visible (figure 2(b)). Incubation of the adsorbed MTs with buffer of high ionic strength resulted in MT removal (figure 2(c)).

## 3.4. Preferred MT orientation by flow adsorption

To increase the yield of MTs positioned parallel to the electrodes a flow adsorption method was used. Thereby, droplets of assembly solution (MTs with MAP2) were repeatedly applied to the tilted substrates (inset, figure 3(a)). They moved along the substrate surface in parts of a second, and left it afterwards. For comparison, droplets were also applied onto substrates which were laying flat, and incubated there for some minutes (inset, figure 3(b)). Samples prepared according to both methods were then washed and air-dried prior to SFM imaging.

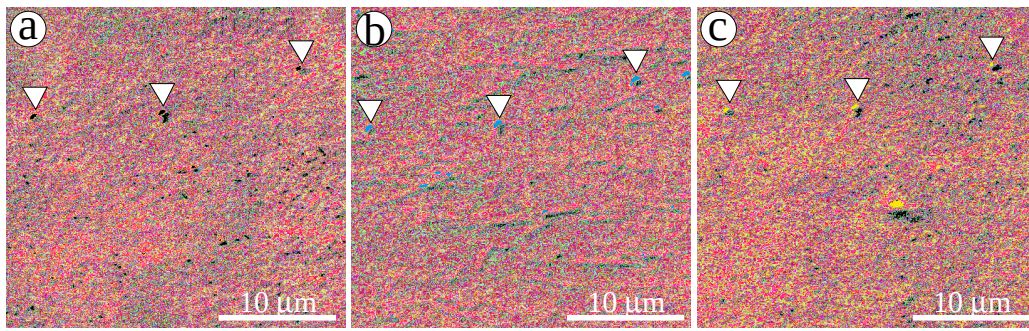
Both procedures yielded samples covered with MTs (figure 3). For characterization of the MT-orientation the directions of all MTs (as given by the long axis) were determined. In the case of flow adsorption, the frequency distribution revealed a clear peak at about  $90^\circ$ , which agrees with the flow direction (from bottom to top, plot in figure 3(a)). The effect of flow is clearly visible in the comparison to the histogram of the control sample, which exhibits a random distribution covering the whole range nearly equally (plot in figure 3(b)). Experiments with different flow directions of the washing fluids and with a variation of the direction of the receding meniscus yielded no differences in the observed orientation, thereby confirming the crucial role of the first adsorption for the MT orientation.

### 3.5. Electron beam-induced deposition of line structures

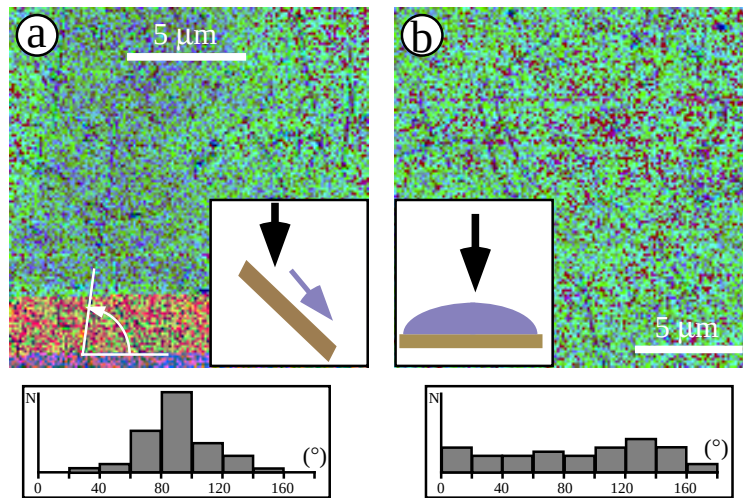
The creation of lines by EBD was tested on the microstructured substrates. Therefore, the slow scan direction of the SEM was disabled. This forces the electron beam to cycle along one scan line, which results in the build-up of a line structure. Some test structures created on silicon oxide between gold electrodes are shown in figure 4(a). Three intersecting lines create a connection through the electrode gap. A typical feature of the EBD lines produced in our SEM is a cone-shaped feature at the beginning of the scan line, with a height clearly exceeding the line height. This effect is probably due to a prolonged duration of the beam in this point compared with the rest of the line, and will be a subject of further studies. The growth rate of the lines is significantly increased on conducting surfaces, e.g. on gold-sputtered surfaces (figure 4(b)). The use of EBD lines for contacting molecules is demonstrated on a rather macroscopic dust particle situated near the electrodes (figure 4(c)). An EBD line connects each electrode with the molecule (figure 4(d)).

### 3.6. Line structures to MTs in electrode gaps

For demonstration of the contacting principle, MT-MAP2 complexes were adsorbed in (or nearby) electrode gaps and visualized by SFM (figure 5(a)). A long MT intersecting one of the electrodes was chosen for further experiments (MT in figure 5(a)). The samples were then transferred to the SEM, and an EBD line was written bridging the gold electrode (E) with the free end of the MT. Figure 5(b) shows the sample in the SEM, the EBD line is faintly visible (arrows).



**Figure 2.** Reversible attachment of MTs formed from pure tubulin to MAP2-covered glass surfaces monitored by video-enhanced DIC microscopy. All images show the same area of the glass surface (cf the three distinct features on the surface, marked by arrowheads). (a) Image of the glass surface after binding of MAP2. (b) Image after inflow of the MT suspension and binding of MTs to MAP2. (c) Image of the glass surface after release of the MTs from the MAP2-covered surface, which was caused by the inflow of buffer with high ionic strength (0.5 M NaCl).



**Figure 3.** The influence of the liquid dynamic onto the adsorption process. MT-MAP complexes were applied in flow (a) from top to bottom or (b) as a sitting droplet. The orientation of the adsorbed MTs was determined from the SFM images and is presented as histogram (bottom). The peak in the distribution for the flow-adsorbed proteins clearly demonstrates the induced orientation compared with the random behavior of the droplet-adsorbed MTs.

### 3.7. Nanoelectrode structuring using EBD masking

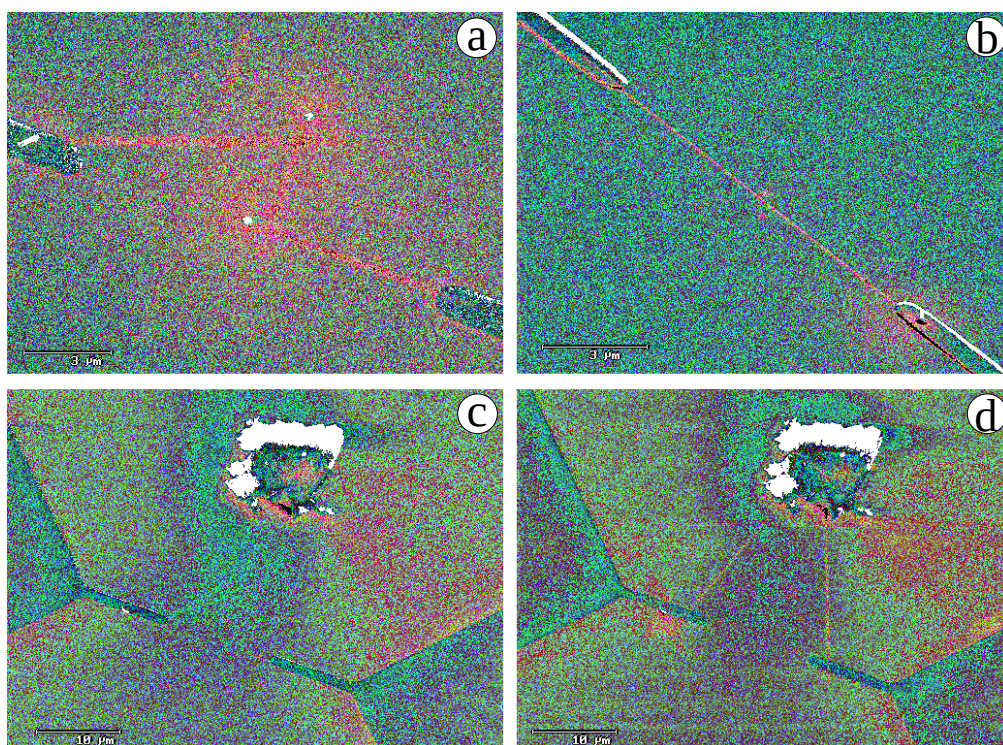
The line structures made by EBD can be transformed into conductive electrodes using a masking technique (figure 6(a)). Therefore, the sample was sputtered with a thin gold film of approximately 30 nm. Then, the EBD line was written using a SEM. In a final step, the gold film was removed by dry etching (with Ar ions). The electrical behavior of electrodes prepared in such way was studied. Therefore, EBD lines were written on a gold-coated substrate between two microelectrodes (figure 6(b)). After dry-etching an  $I-U$ -characteristics of the remaining gold line was measured (figure 6(c)).

## 4. Discussion

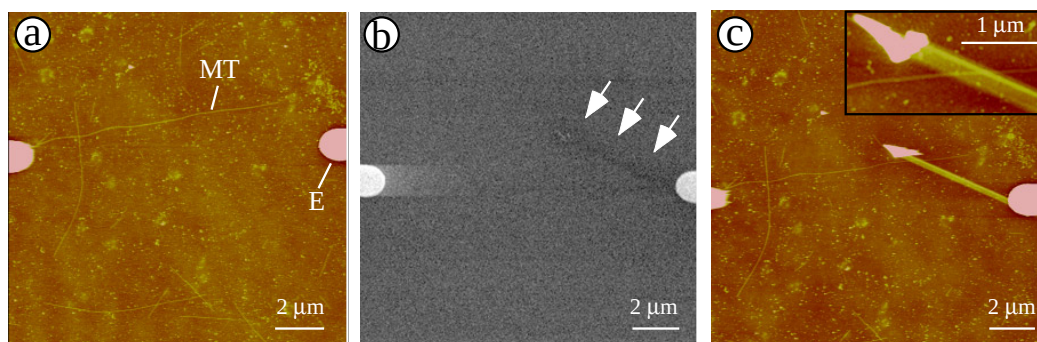
### 4.1. MT adsorption

This paper focuses on steps toward the preparation of electrically contacted biomolecules (namely, MTs) by means of a microstructured surface and electron beam

structuring. Therefore, an adsorption of the MTs to the surface is essential (figure 1(a)). The adsorption of fixed and native MTs onto various substrates (e.g. modified glass, HOPG, modified silicon wafers) has been studied (Fritz *et al* 1995, Hameroff *et al* 1989, Turner *et al* 1995, Vater *et al* 1995, Vinckier *et al* 1995). However, there was no detailed investigation of the materials used for the electrode set-up in the present study: gold and silicon oxide. MTs without MAP2 exhibited no adsorption onto either of these materials. A significant binding of MTs to both materials could be observed after addition of MAP2 to the assembly solution. Apparently, they enable the MT-substrate binding by a bridging effect. For contacting applications, a controlled ad/desorption would be of special interest. Therefore, adsorption studies were conducted by SFM and video-enhanced microscopy using the MAP2-dependent adsorption of MTs. The experiments revealed that MAP2 alone adsorbs to the substrate, and that such a modification is the prerequisite for a high affinity to MTs (in contrast to no adsorption in the case of unmodified glass). The globular structures revealed



**Figure 4.** Nanometer structures by EBD. Scanning electron micrographs. Scale bars are (a, b) 3  $\mu\text{m}$  and (c, d) 10  $\mu\text{m}$ , respectively. (a) Microstructured gold electrodes (broad structures at the left and right) on oxidized silicon were connected by writing three EBD lines. Note the low contrast of the image due to the isolating background. (b) EBD line written and visualized on a conductive background (about 10 nm gold coating). (c, d) A dust particle, located nearby the electrode gap, is used as a test molecule for demonstrating the contacting by an EBD line (d).



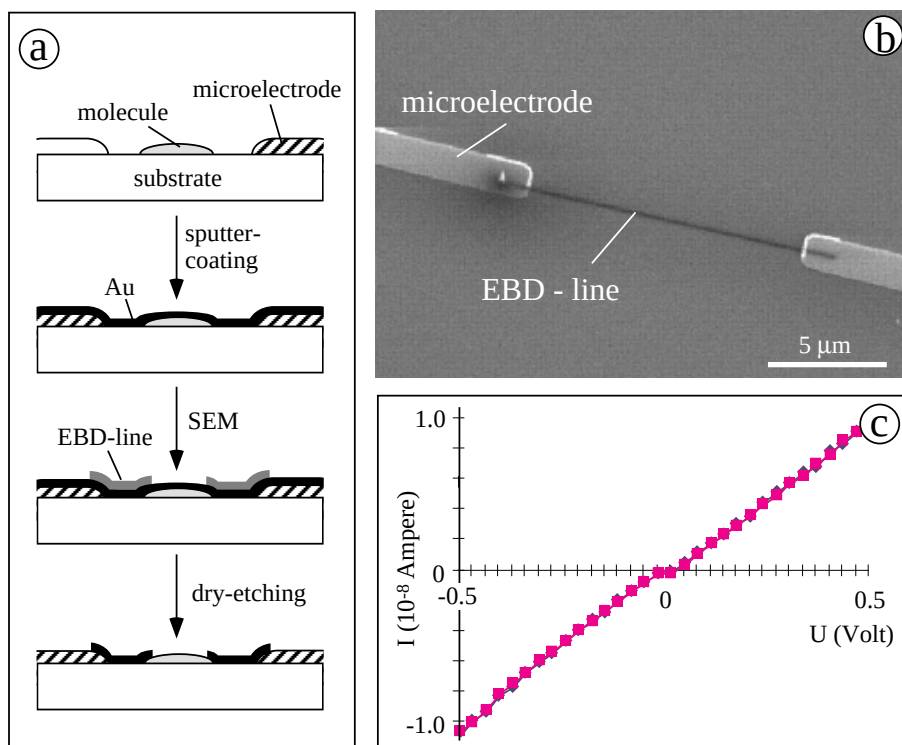
**Figure 5.** Contacting of a single MT. (a) MTs adsorbed in an electrode gap visualized by SFM, the molecule labeled 'MT' was chosen for connection with the right electrode structure ('E'). (b) The SEM was used for the creation and imaging of an EBD line (arrows). The MTs are not visible in the SEM contrast, the structuring was solely based on the SFM image shown in (a). (c) Scanning force micrograph of the connected MT, the written EBD line is clearly visible (cf zoom in inset).

in scanning force micrographs of MAP2-modified surfaces point to aggregates of MAP2. This aggregation effect does not influence the potential to bind MTs, as demonstrated by the adsorption (figure 2(b)) after inflow of MT solution. An interesting feature for controlled deposition of MTs is the reversibility of their MAP2-mediated adsorption. The MAP2-MT interaction is salt dependent, a high ionic strength (as the 0.5 M NaCl used in the experiment) is sufficient to induce a dissociation of the protein-protein complex (Vallee 1983, Vater *et al* 1986). So high-salt buffer removes MAP2-mediated bound MTs, which could be used to remove all molecules which are not contacted after

structuring of the electrodes. In this case, the structured electrodes would arrest the contacted MT on the substrate. An interesting feature in such a set-up is the interface area between MT and substrate. After high-salt conditions, the bridging MAP2 will be removed, and MTs contacted (and therefore fixed) on both ends could be considered as self-supported with minimized substrate interactions.

#### 4.2. MT orientation

For optimization of electrical contacting a parallel orientation of the adsorbed molecules (referred to the



**Figure 6.** Gold wires by EBD-line masking. (a) The contacting scheme is demonstrated: molecules are deposited in the vicinity of microstructured electrodes. EBD lines connecting the molecule with the microelectrodes are written after gold-coating and transfer to a SEM. A dry-etching step removes the gold coating from the surface, resulting in gold wires below the EBD lines. (b) An EBD line between two microelectrodes is used for conductivity tests. (c) Electrical measurement of the same sample as shown in (b) after the dry-etching step.

prestructured gold electrodes) would be helpful. Thereby, the probability of adsorbed molecules which connect both electrodes by spanning over the gap could be increased. Due to the decrease of molecules with other orientations (which disturb electrical measurements in the case of multiple intersections) an increase in MT surface density would be possible (e.g. by increases of adsorption time or MT concentration), resulting in more molecules adsorbed in a desired position. A simple method inducing a preferred MT orientation is the flow adsorption (Turner *et al* 1995). The application of this method to microstructured substrates yielded a high percentage of MTs aligned along the flow direction, and is particularly efficient in the case of longer (5 μm and more) MTs (figure 3).

#### 4.3. MT contacting

Although the results from aligned MTs are promising, there is still the need for a technique which allows the contacting of single molecules out of an ensemble deposited at the substrate surface, e.g. for adjustment of various electrode gaps, or to single-out specified molecules for contacting purposes. The potential of EBD for electrode structuring was investigated. This deposition bases on the effect that the focused electron beam (e.g. in the SEM) induces a build-up of material on the point where it reaches the surface. The material for the created structure comes from the gas inside the vacuum chamber, which results usually in carbonaceous compounds. Electrical characterization yielded a low

conductivity (Fritzsche and Porwol, unpublished results). For the structuring of conducting electrodes the lines are used as etching mask of an underlying conducting layer (figure 6). The successful transformation of such line structures into metal ‘wires’ is demonstrated by the electrical characterization (figure 6(c)). In order to increase the conductivity of this wire, further experiments will aim at an increase of the cross sectional area by thicker gold films and/or by broader EBD lines. On the other hand, it is possible to control the composition of the deposited material by introduction of gases or gas mixtures of defined composition, which can be used for the creation of metal lines (Koops *et al* 1996).

The application of this technique in a SEM allows the structuring but also the immediate control of the created structures (thereby providing a fast feedback). A crucial point is that an orientation using surface features (e.g. gold electrodes) is possible, which allows the contacting of molecules not visible in the SEM by prescreening in a SFM and using the scanning force micrographs. This procedure is explained by figure 5: an electrode structure with adsorbed MTs is visualized by SFM (figure 5(a)). A long MT is chosen for connection to the right electrode (‘E’). This SFM image is the base for the structuring of the EBD line. By using the known angle and distance between the free MT end and the other electrode an EBD line (arrows in figure 4(b)) was written using the SEM in a blind manner. Then, the SFM was used to monitor the results of the EBD (figure 4(c)): a line structure connecting

free MT-end with the free electrode is visible, confirming the successful contact structuring.

## 5. Conclusions

These results demonstrate a great potential for the proposed approach of defined molecule adsorption and subsequent single molecule contacting using the EBD technique. Further work will be aimed toward control of EBD line writing, and the subsequent electrical characterization of the contacted molecules.

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